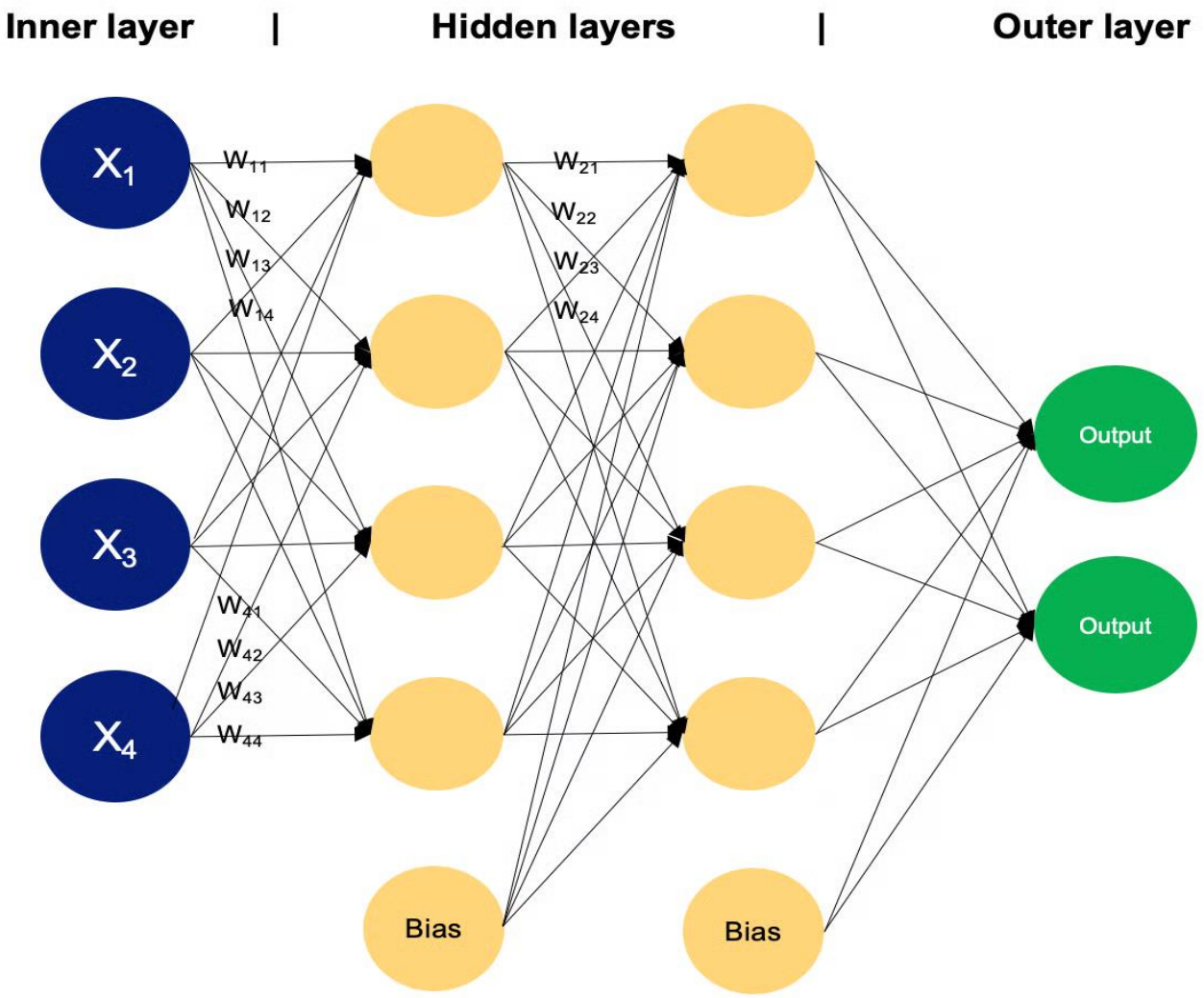
Creating Your First Neural Network in Python with Tensorflow



Creating Your First Neural Network in Python with Tensorflow

steps:

1. Define our training data
2. Create our model
3. Add layers
4. Compile our model
5. Fit our model to our training data
6. Predict values

```
pip install tensorflow --upgrade
```

- Layer type: The type of layer you are creating (ie: Dense, Bidirectional)

- Activation function: A function that modifies the input of a neuron(ie: relu, sigmoid)

- Loss function: A function that measures the inaccuracy of a model in correctly predicting the training data(ie: mean squared error, categorical cross entropy)

- Optimization function: The algorithm that is minimizing the loss value of the model. Is what is actually doing the training(ie: Adaptive Moment Estimation, stochastic gradient descent)

•Loss function: A function that measures the inaccuracy of a model in correctly predicting the training data(ie: mean squared error, categorical cross entropy)

Mean Squared Error (MSE) : One of the most popular loss functions, MSE finds the average of the squared differences between the target and the predicted outputs

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y^{(i)} - \hat{y}^{(i)})^2$$

•Optimization function (Optimizers): The algorithm that is minimizing the loss value of the model. Is what is actually doing the training(ie: adam(Adaptive Moment Estimation), sgd (stochastic gradient descent), Adagrad)

Optimizers are algorithms or methods used to change the attributes of the neural network such as weights to reduce the loss

- Batch Gradient descent
- Stochastic Gradient Descent
- Minibatch Stochastic Gradient Descent
- Adam
- RMSProp
- AdaDelta
- Adagrad

Gradient descent:

When we carry out forward and backpropagation using gradient descent, we can use basically one of three methods:

1. Stochastic gradient descent
2. Mini-batch gradient descent
3. Batch gradient descent

Training Dataset

Subject	Features (Inputs)			Label (Target)
	X1	X2	X3	Y
(1) 1	20	60	44	1
2	15	48	69	0
3	02	59	82	1
4	40	45	74	0

1.If we input **one training example at a time** and carry out forward and backpropagation to update the weights, that is called **Stochastic gradient descent.**

The number of passes = 4 passes (Forward pass and Backward pass) = 1 Epoch.

Training Dataset

(2)

Subject	Features (Inputs)			Label (Target)
	X1	X2	X3	Y
1	20	60	44	1
2	15	48	69	0
3	02	59	82	1
4	40	45	74	0

2. If we input **Multiple training example at a time** (less than the entire training set) that is called **Mini-batch gradient descent**.

The number of passes = 2 passes (Forward pass and Backward pass) = 1 Epoch.

Training Dataset

Subject	Features (Inputs)			Label (Target)
	X1	X2	X3	Y
1	20	60	44	1
2	15	48	69	0
3	02	59	82	1
4	40	45	74	0

(3)

3. If we input **the entire training dataset** and carry out forward and backpropagation, that is called **Batch gradient descent.**

The number of passes = 1 pass (Forward pass and Backward pass) = 1 Epoch.

In Batch gradient descent, we have to process the entire dataset to update the weights, which can be **computationally very expensive** if the dataset is **large**

So, Mini-batch or Stochastic gradient descent is faster than batch gradient descent.

- Most Deep learning algorithms are based on an Optimization algorithm called stochastic gradient descent. It is generally the preferred method for basic Backpropagation.

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Creating and Testing Our Model

Dataset

X1 (Input)	X2 (Input)	Y (Output)
0	0	0
0	1	1
1	0	1
1	1	0

Import the TensorFlow and Keras libraries.

```
Import tensorflow as tf
From tensorflow import keras
From tensorflow.keras import layers
```

Define our training input and output data with type 16 bit float
Each input maps to an output

```
X = tf.constant([[0, 0], [0, 1], [1, 0], [1, 1]], dtype=tf.float16)
Y = tf.constant([[0], [1], [1], [0]], dtype=tf.float16)
```

Create a new Sequential Model

```
model = keras.Sequential ()
```

Add our layers

```
model.add(layers.Dense( 4,        input_dim=2,        activation='relu'))
```

Amount of Neurons .كمية الخلايا العصبية

Define an input dimension because this is the first layer

Use relu activation function

```
model.add(layers.Dense(1,        activation='sigmoid' ))
```

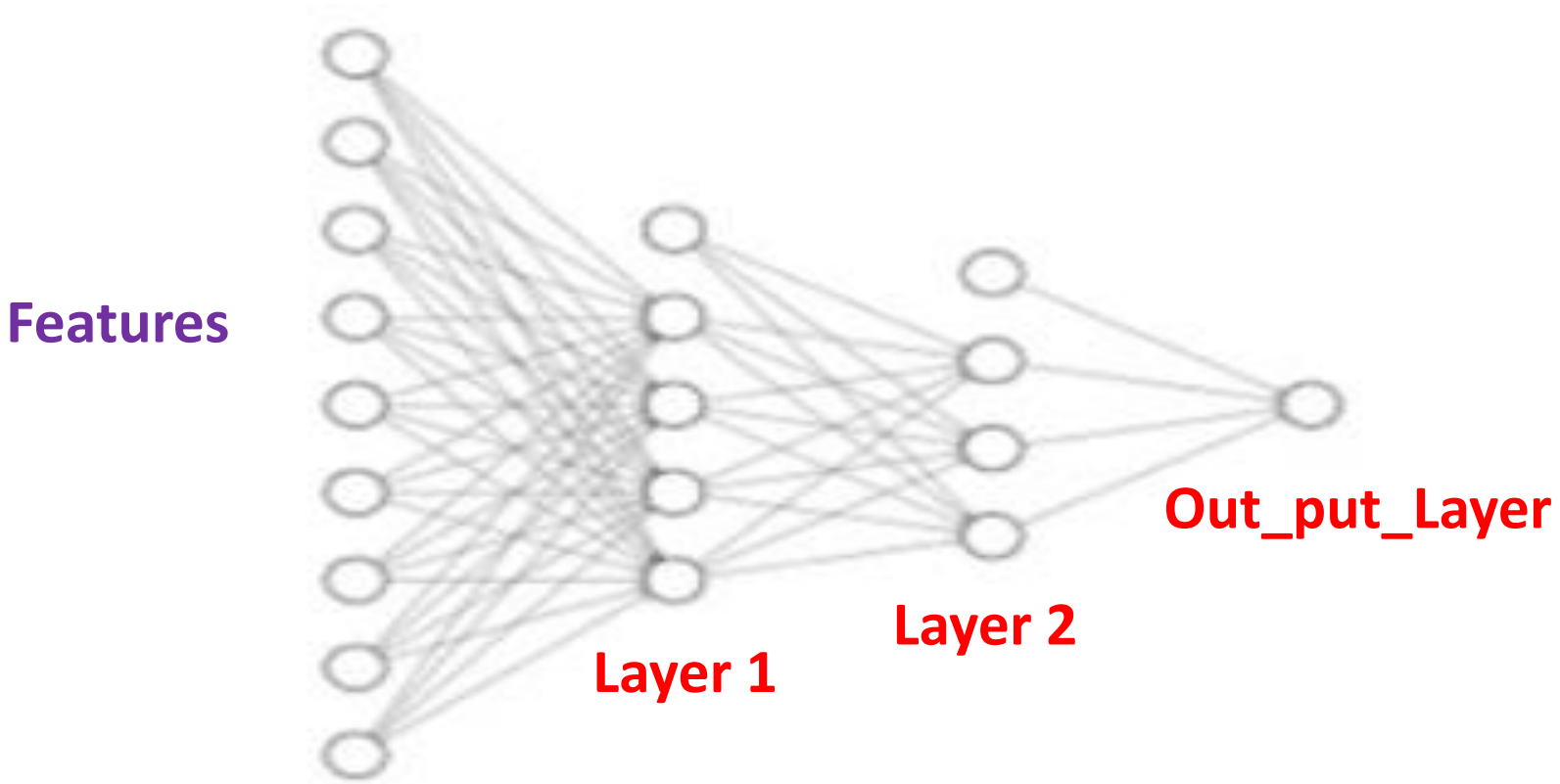
Amount of Neurons. We want one output

Use sigmoid because we want to output a binary classification

Softmax in multiclass

Example of model with 8 features, 4 neurons in layer1, 3 neurons in layer 2 and 1 neuron in output layer

```
model = keras.Sequential ()
model.add(layers.Dense(4, input_dim=8, activation='relu', name='layer 1'))
model.add(layers.Dense(3, activation='relu' , name='layer 2'))
model.add(layers.Dense(1, activation='sigmoid' , name='Output_layer'))
```



Compile our layers into a model

```
model.compile(loss='mean_squared_error', optimizer='adam', metrics=['binary_accuracy'] )
```

The loss function that is being minimized

Our optimization function

Metrics are different values that you want the model to track while training

Fit our model to our training data

```
model.fit(X,        Y,        epochs=200,        verbose=1 )
```

Input training data

Output training data

Amount of iterations we want to train for

Amount of detail you want shown in terminal while training